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import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

# Create sample dataset
app = pd.DataFrame({
    'ID':
    [100001,100002,100003,100004,100005,100006,100007,100008,100009,100010],
    'CNT_CHILDREN': [0,1,0,2,1,0,3,2,1,0],
    'AMT_INCOME_TOTAL':
    [180000,120000,250000,150000,200000,170000,300000,220000,160000,140000],
    'DAYS_BIRTH':
    [-12000,-14000,-10000,-15000,-13000,-11000,-9000,-16000,-12500,-14500],
    'DAYS_EMPLOYED':
    [-3000,-5000,-2500,-6000,-4000,-3500,-1500,-7000,-4500,-5500],
    'FLAG_MOBIL': [1,1,1,1,1,1,1,1,1,1],
    'FLAG_WORK_PHONE': [1,0,1,0,1,0,1,1,0,0],
    'FLAG_PHONE': [1,1,0,1,0,1,0,1,1,0],
    'FLAG_EMAIL': [0,1,0,1,1,0,1,0,1,0],
    'CNT_FAM_MEMBERS': [2,3,2,4,3,2,5,4,3,2]
})

# Display dataset
print(app)

# Correlation Matrix
corr_matrix = app.corr(numeric_only=True)

print("\nCorrelation Matrix:")
print(corr_matrix)

# Plot Heatmap
plt.figure(figsize=(8,6))
sns.heatmap(corr_matrix,
            annot=True,
            cmap='coolwarm',
            fmt='.2f',
            linewidths=0.5)

plt.title("Multivariate Analysis - Correlation Heatmap")
plt.show()

```